HETEROGENEOUS GRAPH NEURAL NETWORKS FOR USER JOURNEY MODELLING: A COMPARATIVE STUDY WITH SEQUENTIAL AND PROBABILISTIC APPROACHES

Dmitrijs Gižbevans

Transport and Telecommunication Institute
Lauvas 2, Riga, LV-1019, Latvia
dmitrijs.gizdevans@gmail.com

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Predicting user behaviour during e-commerce sessions supports real-time personalisation and timely intervention. Unlike post-hoc analysis of completed sessions, prefix-level prediction requires decisions while the session is still in progress. A central challenge in this domain is prefix-level session classification: predicting session outcome from only the first *t* observed events before the session is complete.

Existing approaches to prefix-level behaviour prediction, including probabilistic models such as Markov chains, gradient boosting models on engineered features, and sequential neural architectures such as SASRec (Kang and McAuley, 2018), have limitations for minority-class detection and prediction from short event prefixes. Clickstream data are rich in multirelational dependencies among different types of entities, making graph representations a methodologically well-justified alternative. This study hypothesises that heterogeneous graph models, which explicitly encode entity and relation types (Schlichtkrull et al., 2018), may provide an advantage in prefix-level session classification over sequential and probabilistic approaches, especially for minority classes and short prefixes, where sequential and probabilistic representations may be insufficient.

A series of experiments was conducted on a large-scale e-commerce clickstream dataset with approximately 360,000 sessions and 4.3 million events, covering four behavioural classes (Buyer, Intent, Researcher, Browser) with severe class imbalance (Browser: 88.8%, Buyer: 1.4%). An initial feasibility experiment assessed purchase prediction from static session-level features. Logistic regression and random forest achieved ROC-AUC of 0.997 for completed-session binary classification, indicating substantial predictive signal in the dataset but not capturing sequential dynamics. A sequence-only baseline comparison then evaluated heuristic rules, a first-order Markov chain, and a SASRec transformer under a prefix-level multiclass protocol, achieving test macro-F1 up to 0.42. These results indicated that sequence-only baselines can exploit event order but remain subject to a low performance ceiling. Building on this, an enriched baseline comparison extended the evaluation to Markov chains of orders one through three, gradient boosting on engineered features (Ke et al., 2017), and SASRec ablations incorporating temporal intervals and contextual signals. SHAP-based analysis (Lundberg and Lee, 2017) was used to interpret feature contributions. For Markov models, performance improved consistently with order but remained subject to a clear ceiling. Temporal enrichment provided small but consistent gains for SASRec, whereas contextual features did not yield clear improvement. The best non-graph configuration reached test macro-F1 of 0.58. Across all model families, minority-class performance remained low (Buyer F1 below 0.35), early-prefix prediction remained difficult, and the boundary between Buyer and Intent sessions was consistently ambiguous.

These limitations motivated the application of a heterogeneous graph neural network within the same evaluation framework. The model achieved test macro-F1 of 0.64, with the largest gains on Intent and Buyer.

The baseline experiments showed that sequential and probabilistic models plateau on minority-class detection and early-prefix prediction, even with feature enrichment. The heterogeneous graph model, by explicitly encoding entity and relation types, outperformed the best non-graph baseline by +0.09 macro-F1, with the largest gains on the most difficult classes, Intent and Buyer. These findings suggest that graph-based representations can capture structural signal in clickstream data that sequential and probabilistic approaches fail to model fully.

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References

1. Kang, W. and McAuley, J. (2018) Self-attentive sequential recommendation. In: *Proceedings of the IEEE International Conference on Data Mining (ICDM)*, Singapore, November 2018. IEEE, pp. 197-206.

2. Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q. and Liu, T.-Y. (2017) LightGBM: a highly efficient gradient boosting decision tree. In: *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*. Long Beach: Curran Associates, pp. 3149-3157.

3. Lundberg, S. M. and Lee, S.-I. (2017) A unified approach to interpreting model predictions. In: *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*. Long Beach: Curran Associates, pp. 4765-4774.

4. Schlichtkrull, M., Kipf, T. N., Bloem, P., van den Berg, R., Titov, I. and Welling, M. (2018) Modeling relational data with graph convolutional networks. In: *Proceedings of the European Semantic Web Conference (ESWC)*, Heraklion, June 2018. Cham: Springer, pp. 593-607.